



MAT215: Complex Variables & Laplace Transformations
Quiz-02 (Set-A+B)
July 21, 2026
Total - 20 Marks

Name:

ID:

Section:

Set-A

1. Let

$$f(z) = |z|^2.$$

- (a) Using the definition of the complex derivative, compute $f'(0)$.
- (b) Writing $z = x + iy$, use the Cauchy–Riemann equations to determine all points at which f is complex differentiable.
- (c) Based on your calculations, determine whether f is analytic at $z = 0$.
Hint: Analyticity requires complex differentiability throughout some open neighborhood.

2. Let $f(z) = u(x, y) + iv(x, y)$, where u and v have continuous first-order partial derivatives in a domain D . Let

$$f(z) = \bar{z} e^{-z}.$$

Determine whether u and v have continuous first-order partial derivatives in this case. Find all points at which f is complex differentiable, and compute the derivative wherever it exists. Is the function analytic anywhere?

Set-B

1. Let

$$f(z) = \bar{z}^2.$$

- (a) Using the definition of the complex derivative, compute $f'(0)$.
- (b) Use the Cauchy–Riemann equations to determine all points at which f is complex differentiable.

- (c) Determine whether f is analytic at $z = 0$.

Bonus Question:

1. (a) Let $f(z) = u(x, y) + iv(x, y)$, where u and v have continuous first-order partial derivatives in a domain D . State the necessary and sufficient Cauchy–Riemann conditions for f to be analytic in D .
- (b) Find all the values of $\ln(-3)$.

Best of Luck!